Infosys - Coding 4

A scientific expedition is being prepared for a high-altitude plateau that requires perfect structural symmetry for the transport aircraft. The expedition consists of two distinct groups of specialists: the Vanguard and the Research Team. There are N potential candidates for the Vanguard and M potential candidates for the Research Team.

Each candidate i in the Vanguard has a skill rating V_s[i] and a specialized equipment weight V_w[i]. Similarly, each candidate j in the Research Team has a skill rating R_s[j] and an equipment weight R_w[j].

To ensure the aircraft remains balanced during the flight, the selection must satisfy two strict equilibrium conditions:

1. The total number of specialists selected from the Vanguard must be exactly equal to the total number selected from the Research Team. Let this common number be K.
2. The total weight of the equipment carried by the selected Vanguard members must be exactly equal to the total weight of the equipment carried by the selected Research Team members.

Your objective is to select a valid subset of K candidates from the Vanguard and K candidates from the Research Team that satisfies these equilibrium conditions while maximizing the total combined skill rating of the entire expedition. The total combined skill is defined as the sum of the skill ratings of all chosen Vanguard members plus the sum of the skill ratings of all chosen Research Team members.

If no valid expedition can be formed other than an empty one (i.e., selecting 0 members from each team), the maximum total skill rating is 0.

Find the maximum total combined skill rating achievable.

Input Format

First line: integer N, the number of Vanguard candidates.

Second line: integer M, the number of Research Team candidates.

Next N lines: two space-separated integers per line — skill and weight — for each Vanguard candidate.

Next M lines: two space-separated integers per line — skill and weight — for each Research Team candidate.

Constraints

1 <= N <= 100

1 <= M <= 100

1 <= Vanguard[i][j] <= 1000

1 <= ResearchTeam[i][j] <= 1000

Output Format

The output contains an integer value

Sample Input 0

```
2
2
10 5
20 10
15 5
25 10
```

Sample Output 0

```
70
```

Explanation 0

Selecting both available candidates from the Vanguard and both from the Research Team forms equal groups of 2. Their respective accumulated equipment weights both reach exactly 15, successfully satisfying the equilibrium conditions and combining their skill ratings for a maximum total of 70.

Sample Input 1

```
2
2
100 10
200 20
50 15
150 25
```

Sample Output 1

```
0
```

Explanation 1

Attempting to match any equal number of candidates between the two distinct teams completely fails to produce identical equipment weights. Since no valid balanced groups can be formed to satisfy the structural symmetry, the expedition cannot depart, leaving the maximum total skill at 0.